Ahmedabad University

**Subject** - Digital Image Processing

# 8. Vehicle Orientation Estimation Using Conventional Image Processing

## Weekly Report - 5

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**Repo Link:** https://github.com/encrypted-official/DIP_2026_Project8_Group1

**Group - 1**


| Enrollment No. | Name of the Student | Name of the Program |
|---|---|---|
| AU2440150 | Vehan Gajjar | Btech CSE |
| AU2340012 | Vedant Solanki | Btech CSE |
| AU2440173 | Abhi Jain | Btech CSE |
| AU2440242 | Krish Prajapati | Btech CSE |


2026 (Monsoon Semester)

# 1. Work done this week

In the plan from last week (Table 1 of Weekly Report 4), Week 5 was meant for running Method C and the ablation, first on the 40 image set and then on the full validation split. We changed one thing in the plan. The ablation has 42 settings and each one goes over every crop, so running it on the full split this week was not realistic on our laptops. We used the first 200 validation images instead, which is five times the mid-sem set, and kept the 500 image run and the full split for next week.

**Dataset audit:** Before cutting new crops, we wanted to be sure the labels are read the right way. The new audit_dataset.py goes through every label file of a split, finds the matching image, checks that OpenCV can open it, and parses every vehicle row. It writes the parsed rows to annotation_audit.csv, the image sizes to image_dimensions.csv, and a short summary text file. On the validation split it found 1,184 label files, all with a readable image, 0 missing matches, 0 unreadable images and 52,317 vehicle rows in total. Every image had the same size of 1920 x 1080 pixels. No row was skipped by the parser, so the 11 field format we wrote down in Week 2 still holds, and skipping the first line as flight height is correct.

**Heading direction check:** Last week we said we still had to check whether the dataset measures heading clockwise or anticlockwise. We drew the raw heading as an arrow on 20 crops picked from different classes and checked each one by eye. In all 20 cases the arrow pointed along the body, which means the heading is measured in image coordinates with the y axis pointing down, the same way our estimators measure their angles. So heading mod 180 is the right truth and no sign change is needed.

**Re-running the mid-sem set:** Since the corrected evaluation was ready, we first ran PCA and Hough again on the same 40 images (1822 crops) so that the old and new numbers can be put side by side. Table 1 shows the result.

**Table 1: Mid-sem numbers against the re-run with the corrected angle (40 images, 1822 crops)**

| Metric | PCA (old) | PCA (new) | Hough (old) | Hough (new) |
|---|---|---|---|---|
| MAE (°) | 11.15 | 9.81 | 14.10 | 12.66 |
| Median error (°) | 6.28 | 5.73 | 6.82 | 6.30 |
| P90 error (°) | 27.21 | 24.36 | 39.13 | 34.02 |
| Within 10° (%) | 65.22 | 68.94 | 62.67 | 65.37 |

Both methods got better, which matches what we expected after finding the mirror problem. The biggest change is in P90, around 3 degrees for PCA and 5 for Hough, because the mirrored cases

were mostly sitting in the tail. The ranking stays the same though. PCA is still ahead of Hough on every metric.

**Changes to Method C:** Compared with last week, Method C is now simpler. We dropped the minimum-area rectangle reading and kept only the moments, because on our test images the two readings were within a degree of each other. The other change is in how the blob is chosen. Instead of deciding the vehicle side of the Otsu threshold first, estimate_moments.py now keeps both the bright and the dark mask, cleans both with a 3x3 opening and closing, and collects the connected parts from both. The part whose centre is nearest to the crop centre wins, and if two are equally close the bigger one wins. We also lowered the minimum blob size from 10% to 3% of the crop, since 10% threw away many of the small cars. A blob that is too round (long axis less than 1.15 times the short axis) still gives no prediction.

**First results on 200 images:** We then cut crops from the first 200 validation images. After leaving out difficulty 1 vehicles this gave 9,086 crops. The short side of 1,734 of them is under 20 px, 5,902 are between 20 and 40 px and 1,450 are above 40 px, so a bit under a fifth of the set is in the range where we already knew the methods struggle. All three methods were run with the default settings (blur 3, CLAHE on, Canny 50 and 150 for the two edge methods) and evaluated with evaluate.py. Table 2 gives the numbers.

**Table 2: All three methods with default settings (200 validation images, 9,086 crops)**

| Metric | PCA | Hough | Otsu + moments |
|---|---|---|---|
| Valid / total | 9,075 / 9,086 | 9,068 / 9,086 | 8,612 / 9,086 |
| Coverage (%) | 99.88 | 99.80 | 94.78 |
| MAE (°) | 9.46 | 12.31 | 10.74 |
| Median error (°) | 5.58 | 6.12 | 4.87 |
| P90 error (°) | 23.91 | 33.47 | 27.86 |
| Within 10° (%) | 69.83 | 66.12 | 71.04 |

The PCA and Hough numbers on 200 images are close to the 40 image re-run, so the smaller set was not giving us a lucky picture. Method C is the interesting one. When it gives an answer it has the lowest median (4.87 degrees) and the highest share within 10 degrees, so on a clean crop the blob gives a better axis than the edges. But it skips about 5% of the crops, and its P90 is worse than PCA. When we opened the skipped and the worst crops, most were grey or silver cars on grey road where Otsu had nothing to split, and dark cars whose shadow joined the blob and made it wider on one side. Because the crops it skips are the hard ones, its MAE is computed on an easier set than the other two, and we have to keep that in mind before calling it the best.

**Ablation:** The ablation.py script runs PCA and Hough over blur sizes 1, 3 and 5, CLAHE on and off, and Canny pairs (30, 90), (50, 150) and (80, 200), which is 18 settings each. Method C has no Canny step, so it only gets the 6 blur and CLAHE settings. That makes 42 runs, not 48 as we wrote last week, because Method C now has only one reading. For every run the script saves coverage, MAE, median, P90 and within 10 degrees, plus a per-crop file with every prediction. The full grid on 9,086 crops took about 2 hours 40 minutes on one laptop. Figure 1 shows how MAE moves with blur size, and Table 3 lists the setting we picked for each method.

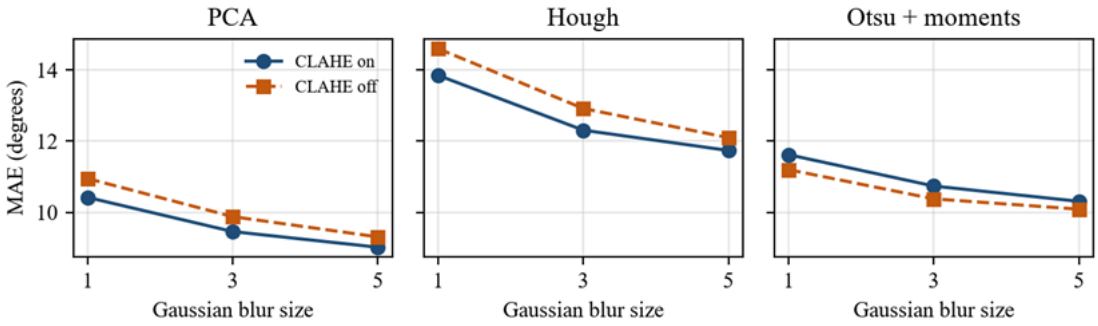



**Figure 1: MAE against blur size with CLAHE on and off (Canny 50, 150 for PCA and Hough)**

**Table 3: Setting picked for each method from the ablation (9,086 crops)**

| Method | Blur | CLAHE | Canny | Cov. (%) | MAE (°) | Median (°) | P90 (°) | ≤10° (%) |
|---|---|---|---|---|---|---|---|---|
| PCA | 5 | On | 80, 200 | 99.52 | 8.73 | 5.21 | 21.64 | 71.92 |
| Hough | 3 | On | 80, 200 | 99.31 | 11.08 | 5.94 | 29.85 | 67.88 |
| Otsu + moments | 5 | Off | N/A | 95.41 | 10.09 | 4.71 | 25.92 | 72.36 |

A few patterns came out of the grid. More blur helped all three methods. Going from blur 1 to blur 5 cut the MAE by about 1.4 degrees for PCA and 2.1 for Hough, since the extra smoothing removes the small texture edges on the roof and the road that were pulling the axis sideways. The higher Canny pair (80, 200) also helped for the same reason. CLAHE helped the two edge methods by a small amount, but it made Method C slightly worse. Our guess is that CLAHE raises the contrast of shadows as well, so the shadow gets cut out together with the car more often.

We did not just take the lowest MAE from the grid. For Hough the lowest MAE was 10.62 degrees at blur 5, CLAHE off and (80, 200), but at that setting coverage fell to 96.9%, because on small crops the strong smoothing leaves no line long enough to pass the length check. We picked the second best instead, which keeps coverage above 99%. For PCA the best setting also

had good coverage, so there was no such trade-off.

**Error by crop size:** Figure 2 splits the MAE by crop size at default settings. All three methods are fine above 40 px and even between 20 and 40 px the error stays under 10 degrees. Below 20 px everything breaks, with MAE between 25 and 31 degrees, and Method C also skips around 19% of these crops. No preprocessing setting in the grid fixed this, so the small crops will need a separate look in Week 7.

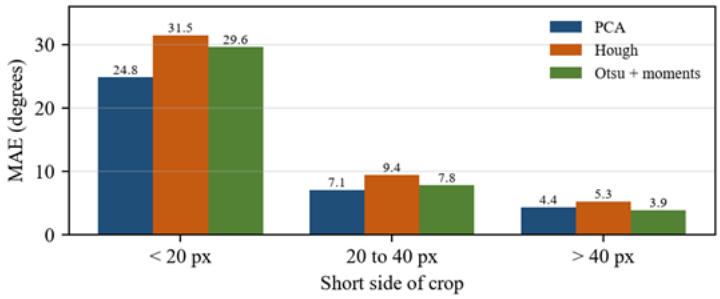



**Figure 2: MAE by crop size for the three methods (default settings, 9,086 crops)**

**Integration fixes:** Putting the four parts together was not as smooth as copying files. Three things broke. First, the new angle_utils.py no longer had the heading_to_axis180 function, so evaluate.py stopped importing. We put it back. Second, ablation.py read the truth from the gt_axis_deg column of metadata.csv, and extract_crops.py still writes that column with the old 0 to 90 fold. This would have brought back the same mirror problem we fixed last week, so the ablation now builds the truth from heading_deg mod 180, the same as evaluate.py. Third, test_ablation.py imported a build_configurations function that the ablation script did not have, so we changed the test to check the grid lists directly. After these fixes all 24 unit tests pass, including the new mirror case where 135 and 45 degrees must be 90 degrees apart.

## 2. Work to be done next week

Week 6 in our plan is front vs back heading, but we also need to finish the bigger runs. We plan to:

- Cut crops from 500 validation images and run all three methods with the settings from Table 3, then move to the full validation split if the numbers stay stable.
- Report Method C both on its own valid crops and on the crops all three methods answered, so that the comparison is fair.
- Try a simple shadow check for Method C, for example dropping the darkest part of the blob when it sits on one side only.
- Start the front vs back step by looking at the intensity profile along the predicted axis, since the windscreen and the rear window usually look different.

## 3. Literature Review

This week was mostly about preprocessing settings, so we read about the steps the ablation was changing.

- Canny (1986) set out the three goals of an edge detector: good detection, good localisation and a single response per edge. The paper also shows that the Gaussian width trades noise against detail, which is what we saw when larger blur sizes removed roof texture but also removed edges on the smallest cars.
- Lindeberg (1998) argues that there is no single right smoothing scale and that it should depend on the size of the structure. A blur that depends on crop size could help our small cars.
- Pizer et al. (1987) and Zuiderveld (1994) describe adaptive histogram equalisation and the clip limit that keeps CLAHE from boosting noise.
- Matas, Galambos and Kittler (2000) describe the progressive probabilistic Hough transform that OpenCV's HoughLinesP is based on. Lines are found from random samples of edge points, which is why the minimum length and the vote threshold matter so much on small crops with few edge pixels.
- Mardia and Jupp (2000) cover statistics for angles and axes. Doubling the angle before averaging, which we use when combining Hough lines, is the standard way to average axial data where 10 and 190 degrees are the same direction.

## 4. Challenges Faced

- Time. Each ablation run goes through every crop, and 42 runs on 9,086 crops took hours. We had to leave a laptop running overnight and could not try the full split this week.
- Making the Method C comparison fair. It skips the hardest crops, so its MAE looks better than it should. We still have to agree on how to report this.
- Vehicles under 20 px. No setting in the grid helped them much, and they make up almost a fifth of the crops.

## References


[1] J. Canny, "A Computational Approach to Edge Detection," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. PAMI-8, no. 6, pp. 679-698, 1986.

[2] T. Lindeberg, "Edge Detection and Ridge Detection with Automatic Scale Selection," International Journal of Computer Vision, vol. 30, no. 2, pp. 117-156, 1998.

[3] S. M. Pizer et al., "Adaptive Histogram Equalization and Its Variations," Computer Vision, Graphics, and Image Processing, vol. 39, no. 3, pp. 355-368, 1987.

[4] K. Zuiderveld, "Contrast Limited Adaptive Histogram Equalization," in Graphics Gems IV, P. S. Heckbert, Ed. Academic Press, 1994, pp. 474-485.

[5] J. Matas, C. Galambos and J. Kittler, "Robust Detection of Lines Using the Progressive Probabilistic Hough Transform," Computer Vision and Image Understanding, vol. 78, no. 1, pp. 119-137, 2000.

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[7] DRASHTI-HaOBB dataset, Zenodo, https://zenodo.org/records/18278989